

Hidden States

$$Z_{in} \in \{C, H, N\}$$

i = game index

n = shot number

Z_{in} = hidden state during shot n

Y_{in} = observed shot outcome, made or missed $\in \{0, 1\}$

Transition Matrix

Stores probabilities for going from state (C, H, N) to state (C, H, N) .

Row = current state

Column = next state

Looks like

$$P = \begin{pmatrix} p(CC) & p(CN) & p(CH) \\ p(NC) & p(NN) & p(NH) \\ p(HC) & p(HN) & p(HH) \end{pmatrix}$$

and each row should sum to 1.

Transition dist.

$$Z_{i,n+1} | Z_{in} \sim \text{Categorical}(p_i(j|C), p_i(j|N), p_i(j|H)), \quad j \in \{C, H, N\}$$

Softmax regression

Used for state transition probabilities:

$$j \in \{C, H, N\}$$

$$\eta_i(j|C) = X_i \beta(j|C) + b_i(j|C)$$

$$\eta_i(j|H) = X_i \beta(j|H) + b_i(j|H)$$

$$\left. \begin{array}{l} \eta_i(j|C) = X_i \beta(j|C) + b_i(j|C) \\ \eta_i(j|H) = X_i \beta(j|H) + b_i(j|H) \end{array} \right\} \begin{cases} p_i(j|C) = \frac{e^{\eta_i(j|C)}}{1 + e^{\eta_i(j|C)} + e^{\eta_i(j|H)}} \\ p_i(j|H) = \frac{e^{\eta_i(j|H)}}{1 + e^{\eta_i(j|C)} + e^{\eta_i(j|H)}} \\ p_i(j|N) = \frac{1}{1 + e^{\eta_i(j|C)} + e^{\eta_i(j|H)}} \end{cases}$$

X_i = vector of features from dataset

Ex.
$$\begin{bmatrix} \text{shot distance} \\ \text{time remaining} \\ \text{fatigue} \\ \text{defender distance} \end{bmatrix} = \begin{bmatrix} 1 \\ 22.5 \\ 180 \\ 0.7 \\ 3.2 \end{bmatrix}$$
 dummy values

β = global coefficients for these features

Same size as X_i

Learned probabilities by the model so that the transition probabilities match observed data well. [...] form

b_i = how much a game deviates from average

Given its game i do we know if the player will be more streaky if so,

$b_i(jH) = +0.8 \rightarrow$ increase probability of transitioning to hot

Initial Distribution

$$\delta_i = (\delta_i(C), \delta_i(N), \delta_i(H))$$

Starting with $\delta_i \sim \text{Dirichlet}(1, 1, 1)$

So all states equally likely at start of game i .

If a player is more likely to start cold adjust to $\text{Dirichlet}(5, 1, 1)$ so cold state is higher prob.

Shot Outcome Model

$$Y_{in} | Z_{in} = C \sim \text{Bernoulli}(\gamma_{in}(C))$$

$$Y_{in} | Z_{in} = N \sim \text{Bernoulli}(\gamma_{in}(N))$$

$$Y_{in} | Z_{in} = H \sim \text{Bernoulli}(\gamma_{in}(H))$$

$Y_{in} = 1$ if made

$Y_{in} = 0$ if missed

γ_{in} is the success probability in state

Shot prob. regression

$$\text{logit}(\gamma_i(c)) = X_i \alpha(c) + \sum a_i(c)$$

X_i = design matrix of observed features in game i .

Ex.

$$X_i = \begin{bmatrix} 1 & \text{distance}_1 & \text{time}_1 \\ 1 & \text{distance}_2 & \text{time}_2 \\ 1 & \text{distance}_3 & \text{time}_3 \end{bmatrix}$$

where each row is a shot

$\alpha(c)$ = fixed-effect coefficients